GlucoTwin

A virtual patient workspace for glucose forecasting

Historical clinical context + wearable sensor history

Research prototype

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Problem and healthcare use case

A clinician needs to see how recent glucose changes relate to a patient's historical clinical context.

GlucoTwin links the profile to visible sensor history and fixed future forecasts.

The research objective is advance prediction of a new sustained high-glucose episode.

Linked historical and sensor data

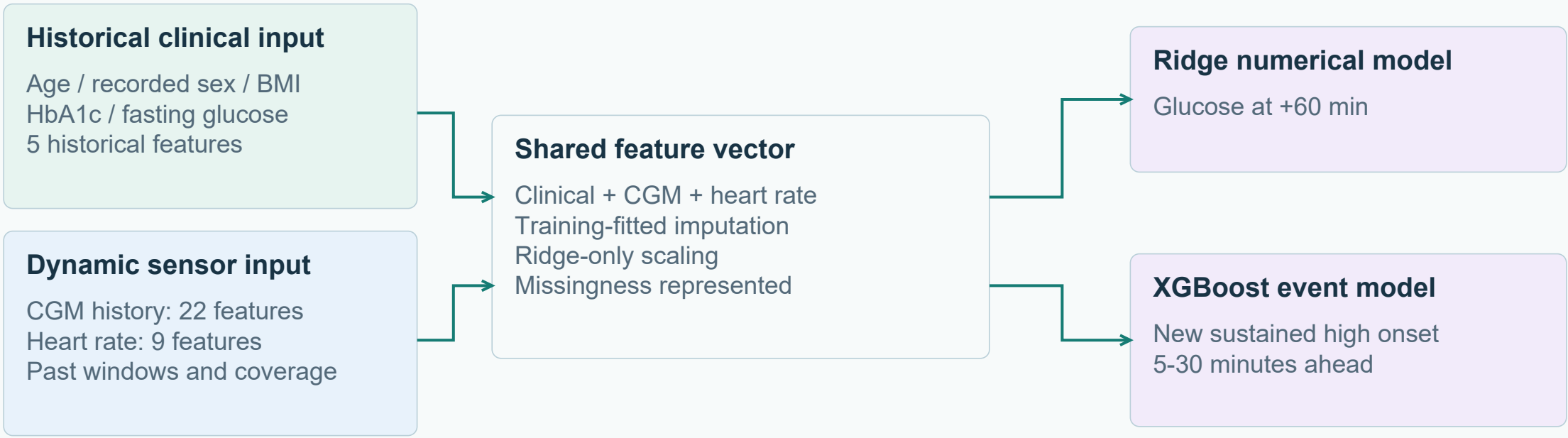
CGMacros is the nutrition and diet-monitoring cohort. Historical inputs: age, recorded sex, body mass index (BMI), HbA1c and fasting glucose.

Dynamic input: continuous glucose monitoring (CGM) and heart rate (HR), linked to the same patient. RBG refers to the REPLACE-BG study.

Retrospective replay; CGM samples are reconstructed and clinical availability is assumed.

Clinical and sensor fusion

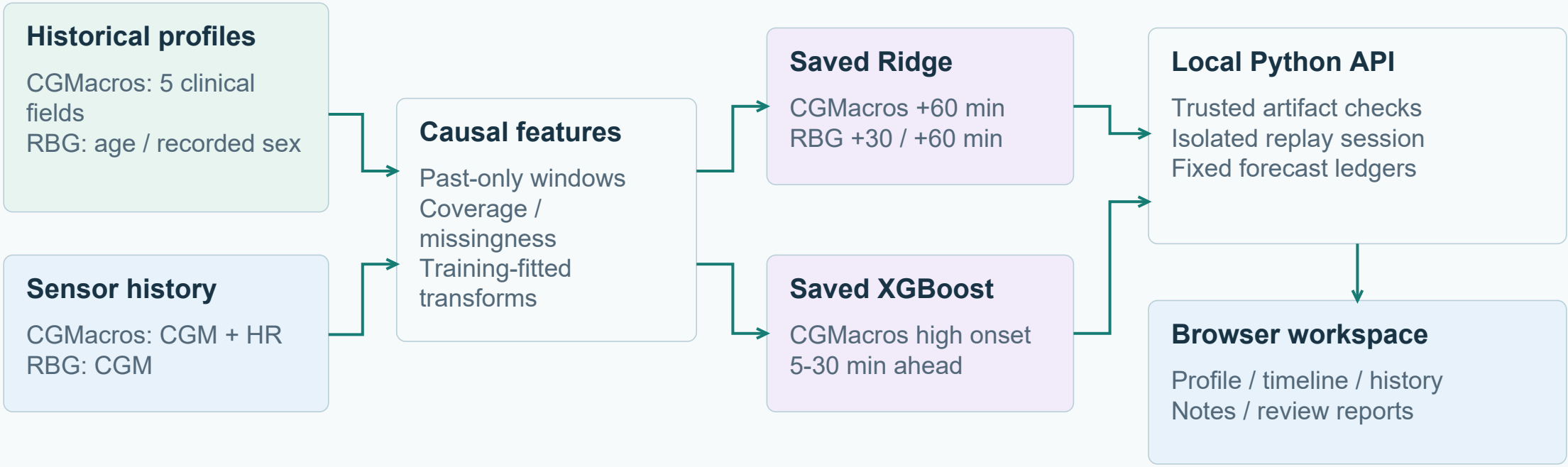
CGMacros links both streams by patient ID; historical profile availability is assumed.



Fusion is implemented. A useful predictive advantage from clinical inputs has not been established.

System architecture

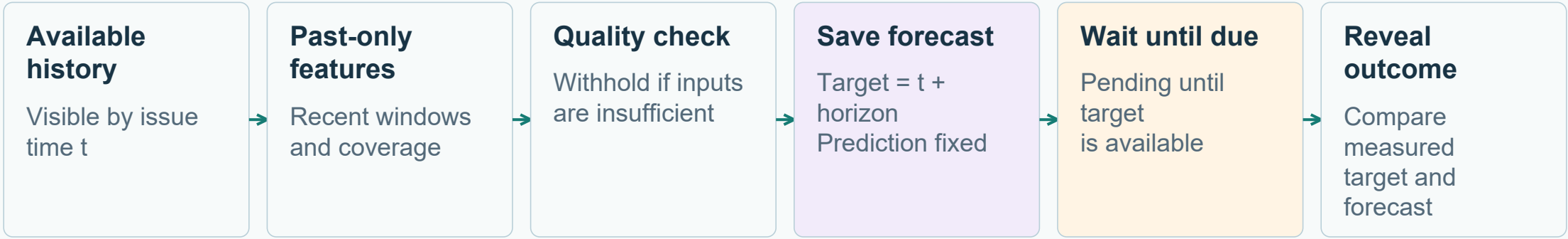
Linked profiles and available sensors enter fitted models through causal features.



Source records and models stay local. Public code, protocols and aggregate results are shared.

Forecast timing and outcome reveal

Inputs are past-only; predictions stay fixed and future target readings remain hidden.



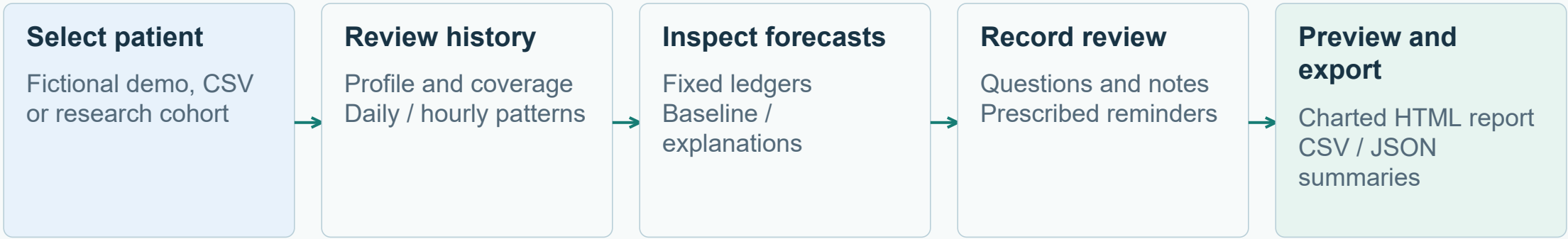
Availability rules

CGMacros: input cutoff t; target revealed at t + horizon
RBG: input cutoff t - 5 min; target revealed at t + horizon + 5 min

Withheld forecasts are not evaluated. Missing targets stay missing. Event research has a separate +45-minute outcome rule.

Clinician review workflow

Inspect available evidence, record questions and retain user-entered follow-up tasks.



Available in the current build

Synthetic worked example / Local CSV import / Sensor gap inspection
Observed day comparisons / User-entered plans / Printable review reports

Notes do not enter inference. The three-day plan is user-entered; medication times and doses are not predicted.

Advance-event definition

New sustained high onset: 4 consecutive five-minute readings above 180 mg/dL, after 7 readings at or below 180.

Prediction window: onset 5-30 minutes ahead. Features contain only past information.

27 development patients; 3 outer folds separate fitting, cutoff setting and evaluation. Six reserved patients remain unscored.

Measured event prediction

Metric	Fused XGBoost v2
Episodes detected	229 / 336 (68.2%)
Conservative precision	52.8%
False alarms / eligible day	0.98
Median detected-event lead	10 minutes

Development results. Cutoff-setting folds were not consistently successful; patient intervals cross gate limits.

Numerical forecast results

Cohort / horizon	Ridge MAE	Persistence MAE
CGMacros / 60 min	19.00	19.99
RBG / 30 min	17.59	20.71
RBG / 60 min	28.89	32.53

Units: mg/dL. Separate development cohorts and timing protocols. RBG profile-fused values match the original CGM-only benchmark after rounding.

Current model limitations

Only 36.6% of evaluable high episodes received at least ten minutes warning. Cutoff-setting folds did not consistently pass.

Low-glucose numerical errors remain a weakness. Overall MAE is not a clinical safety guarantee.

CGMacros samples are reconstructed; historical profile availability is assumed. A useful clinical-fusion advantage is not established.

No clinical validation or reserved-test performance. Risk percentages, operational alerts and treatment advice remain disabled.

Setup and reproducibility

Python + scikit-learn + XGBoost; HTML/CSS/JavaScript browser. Measured studies ran on local CPU.

Synthetic demo runs without data or model downloads. Research cohorts require locally prepared, trusted artifacts; raw records and models stay local.

One-click Windows launcher; unified Python and JavaScript checks. CSV import, care notes, history inspection and charted review reports are included.

Original code: MIT License. Dataset terms remain separate. Video preparation is deferred.